ERICK ALBERTO SÁNCHEZ ARANDA

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EDUCATION

Tecnológico de Monterrey (ITESM) **Aug. 2024 – Expected Jun. 2028**

B.S. in Computer Science and Technology | GPA: 9.0/10.0 *Guadalajara, Jalisco*

- **Coursework:** Data Structures and Algorithms, Computer Architecture, Computer Networks, IoT Software Engineering, Cybersecurity
- **Planned concentration:** Semiconductors (E1002–E1004), starting Aug. 2027

EXPERIENCE

Undergraduate Research Assistant, Water Filtration Telemetry **Aug. 2026 – Present**

Tecnológico de Monterrey *github.com/uno21858/water-column-telemetry*

- Building the telemetry stack for a four-column water filtration rig: interrupt-driven ESP8266 firmware in C++ that counts flow-sensor pulses per column, a Supabase (PostgreSQL) schema for the readings, and a Next.js dashboard deployed on Vercel.
- Chose interrupt-driven pulse counting over polling for the four flow sensors: ISRs placed in IRAM, as the ESP8266 requires, and shared counters read inside critical sections so no pulse is lost or torn between the ISR and the main loop.

Undergraduate Research Assistant, Computer Vision **Feb. 2026 – Jun. 2026**

Tecnológico de Monterrey | Advisor: Prof. Atoany Fierro Radilla *github.com/uno21858/traffic-sign-cnn*

- Trained a custom ~628K-parameter CNN in PyTorch on GTSRB traffic signs (43 classes), reaching 92% on the official test set.
- Set up GPU training on a self-hosted RHEL server (Docker, NVIDIA Container Toolkit) and served the model with FastAPI.

Administrative Assistant (Part-time) **Aug. 2022 – Present**

Transportes Sánchez Barcelata *Guadalajara, Jalisco*

- Built FDM (Factura Diesel Manager) on own initiative: a Python desktop app (PySide6, Selenium) that downloads diesel invoices from Mexico's tax authority (SAT) and totals liters and cost, replacing a manual reconciliation workflow; ships as a Windows installer (PyInstaller + NSIS).
- Adapted FDM's architecture into a bulk payroll-invoice downloader (PDF + XML) with batch scheduling and retry logic to pull 5,000+ invoices for a full fiscal-year compliance audit within SAT's daily download limits.

LEADERSHIP & ACTIVITIES

PawnGuard, Cybersecurity Student Group **Oct. 2025 – Present**

Treasurer, previously Infrastructure Staff *Guadalajara, Jalisco*

- Co-organized Hack2Dawn, an internal jeopardy-style CTF sponsored by MLH, 330ohms and Google Cloud; engineered its CI/CD pipeline (GitHub Actions, Docker) that provisions CTFd on a VPS and monitored the server during the live event.
- Developed two OSINT challenges for the CTF: GitHub commit forensics and search-engine dorking for exposed documents.

PROJECTS

Home Water SCADA | *C++, ESP32-S3, ESP-IDF, LVGL, KiCad* **Sep. 2026 – Present**

Monitoring and chlorination controller for a home water system | self-directed *github.com/uno21858/home-water-scada*

- ESP32-S3 controller with a 7-inch touch HMI reading a hall-effect flow sensor, two pressure transducers via a 16-bit I2C ADC (ADS1115) and a 4-20 mA level transmitter, plus a MOSFET-driven 12 V mixing pump; meters water volume to compute chlorine dosing.
- Firmware on ESP-IDF with FreeRTOS tasks (sensors, LVGL, uploads), signed OTA updates over WiFi with automatic rollback, and a telemetry client (Influx line protocol, one-day offline queue) for a home metrics server; HMI in C++ with LVGL, developed on a desktop simulator before the board.
- Designed the sensor node in KiCad (12 V input, 5 V regulation, 24 V boost, fused and zener-clamped input, flyback diode), built it on perfboard and bench-tested every channel, expected vs. measured, before integration; measured values recorded in the repo.

Cuentas Claras LATAM | *Flutter, Supabase, AWS* **2025 – Present**

Expense-splitting mobile app | solo developer *cuentasclaraslatam.com*

- Shipped to the App Store and Google Play. The client never queries tables directly: ~170 SECURITY DEFINER RPCs that check the caller's Cognito JWT, RLS on all 21 tables, bank details encrypted with pgcrypto and a key held in Supabase Vault.
- Security-tested my own production API: intercepted Flutter traffic with reFlutter, Frida/objection and Burp Suite, replayed requests with a second account's JWT, and found and fixed an IDOR that let a regular user modify other users' data.

LLM Fine-Tuning | *PyTorch, PEFT, Ollama* **2026**

Self-directed | self-hosted RHEL server with two RTX 3060 GPUs

- Fine-tuned Qwen3-8B with QLoRA on 200K+ personal chat messages to model a conversational persona; merged adapters with PEFT, quantized to GGUF and served it locally via Ollama.

Zenalyse | *Raspberry Pi, AWS, Next.js* **Nov. 2025**

IoT environmental monitoring | 1st Place out of 8 engineering teams, Skilliket Symposium

- Backend and project manager in a 6-person team: AWS EC2 backend with PostgreSQL behind a Next.js dashboard, correlating Raspberry Pi sensor data (DHT11, MQ-135, PIR) with user stress surveys.

SKILLS

Programming: C++, Python, Dart, SQL (PL/pgSQL), TypeScript/JavaScript, Java, Bash
Embedded: ESP32-S3 / ESP8266 (ESP-IDF, FreeRTOS tasks, PlatformIO), ARM Cortex-M33 (RP2350 / Pico 2 W, Pico SDK), interrupts and ISRs, I2C, UART, ADC and 4-20 mA sensor interfacing, LVGL, KiCad, bench testing with a multimeter, 3D printing
ML & Data: PyTorch, PEFT (QLoRA), NumPy, Pandas, scikit-learn, CUDA (basics)
Backend & Cloud: Supabase, PostgreSQL, AWS (Cognito, Lambda, EC2), Firebase, FastAPI
Frameworks & Tools: Flutter, Next.js, PySide6, Selenium, Docker, Git, GitHub Actions, Linux (RHEL, Ubuntu), Cloudflare (Zero Trust, Tunnels)
Security: Burp Suite, Frida, objection, reFlutter, Nmap, Wireshark
Certifications: AWS Certified Cloud Practitioner (Aug. 2025), Oracle Java Foundations (2023)
Languages: Spanish (native), English (professional working proficiency, Cambridge B2 First)